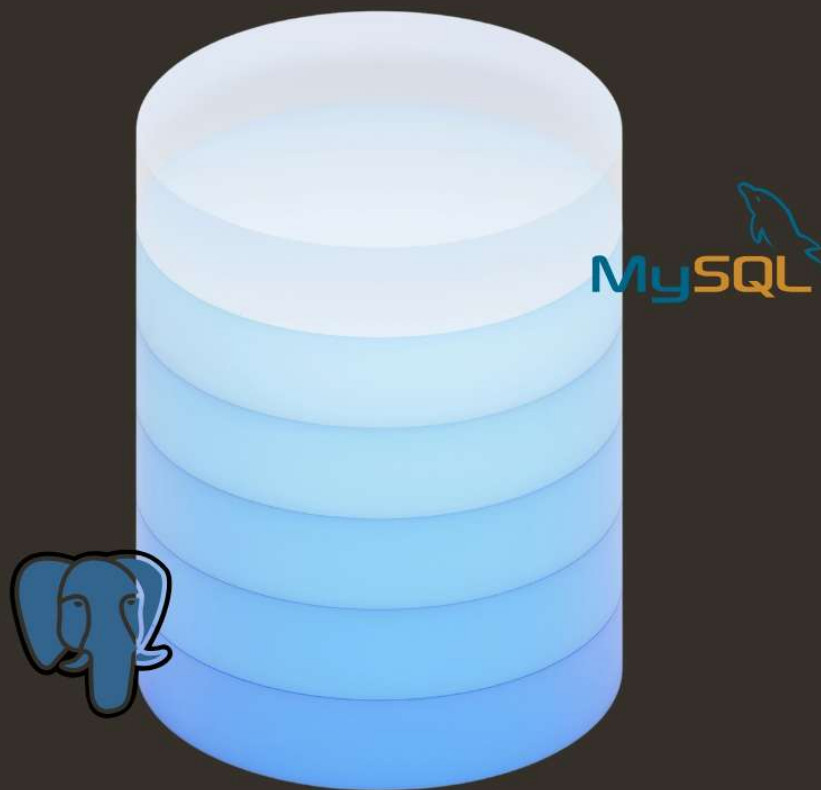


SQL EXAM QUESTIONS



SQL Exam MCQS

1. SQL stands for.
 - A. Structured Query Language
 - B. Simple Query Language
 - C. System Query Language
 - D. Sequential Query Language

Answer: A. Structured Query Language

Explanation: SQL stands for Structured Query Language and is used to communicate with relational database systems.

2. Which of the following is a popular relational database management system?
 - A. MySQL
 - B. Photoshop
 - C. HTML
 - D. Git

Answer: A. MySQL

Explanation: MySQL is an RDBMS that stores data in related tables and supports SQL for managing and querying data.

3. Which SQL command is used to create a new database?
 - A. MAKE DATABASE
 - B. NEW DATABASE
 - C. CREATE DATABASE
 - D. BUILD DATABASE

Answer: C. CREATE DATABASE

Explanation: CREATE DATABASE database_name; creates a new database in MySQL.

4. Which command is used to select a database for use in MySQL?
 - A. SELECT DATABASE
 - B. USE
 - C. OPEN
 - D. CONNECT

Answer: B. USE

Explanation: The USE command makes a particular database the active database for subsequent operations.

5. Which SQL statement is used to create a table?

- A. BUILD TABLE
- B. CREATE TABLE
- C. NEW TABLE
- D. MAKE TABLE

Answer: B. CREATE TABLE

Explanation: CREATE TABLE defines a new table along with its columns, data types, and constraints.

6. Which data type is most appropriate for storing whole numbers?

- A. VARCHAR
- B. DATE
- C. INT
- D. DECIMAL

Answer: C. INT

Explanation: INT is commonly used to store integer values such as age, quantity, employee ID, or number of products.

7. Which MySQL data type is commonly used to store variable-length text?

- A. INT
- B. VARCHAR
- C. DATE
- D. BOOLEAN

Answer: B. VARCHAR

Explanation: VARCHAR stores variable-length character strings and is commonly used for names, cities, email addresses, and categories.

8. Which data type is suitable for storing a calendar date?

- A. DATE
- B. INT
- C. VARCHAR
- D. FLOAT

Answer: A. DATE

Explanation: The DATE data type stores date values such as 2026-08-25.

9. Which constraint uniquely identifies each record in a table?

- A. UNIQUE
- B. NOT NULL
- C. PRIMARY KEY
- D. DEFAULT

Answer: C. PRIMARY KEY

Explanation: A primary key uniquely identifies each row in a table and cannot contain NULL values.

10. Which constraint prevents a column from containing NULL values?

- A. UNIQUE
- B. NOT NULL
- C. DEFAULT
- D. CHECK

Answer: B. NOT NULL

Explanation: NOT NULL ensures that value must be provided for the column when a record is inserted.

11. Which constraint prevents duplicate values in a column?

- A. UNIQUE
- B. DEFAULT
- C. NOT NULL
- D. INDEX

Answer: A. UNIQUE

Explanation: The UNIQUE constraint ensures that values in the specified column are not duplicated.

12. What is the purpose of the DEFAULT constraint?

- A. To delete duplicate records
- B. To automatically provide a value when none is supplied
- C. To make a column a primary key
- D. To prevent NULL values

Answer: B. To automatically provide a value when none is supplied

Explanation: A DEFAULT value is automatically inserted when an INSERT statement does not provide a value for that column.

13. Which command is used to add a new column to an existing table?

- A. UPDATE TABLE
- B. ALTER TABLE
- C. MODIFY TABLE
- D. CHANGE DATABASE

Answer: B. ALTER TABLE

Explanation: ALTER TABLE is used to modify the structure of an existing table, including adding or removing columns.

14. Which statement correctly adds an email column to a Customers table?

- A. ADD TABLE Customers email VARCHAR(100);
- B. ALTER TABLE Customers ADD email VARCHAR(100);
- C. UPDATE Customers ADD email VARCHAR(100);
- D. CREATE COLUMN email VARCHAR(100);

Answer: B. ALTER TABLE Customers ADD email VARCHAR(100);

Explanation: The ADD operation with ALTER TABLE adds a new column to an existing table.

15. Which SQL command is used to permanently remove a table and its structure?

- A. DELETE
- B. REMOVE
- C. DROP TABLE
- D. TRUNCATE TABLE

Answer: C. DROP TABLE

Explanation: DROP TABLE removes the table itself, including its structure and stored data.

16. What does the following command do?

TRUNCATE TABLE Students;

- A. Deletes the database
- B. Removes all rows while keeping the table structure
- C. Deletes only one row
- D. Removes one column

Answer: B. Removes all rows while keeping the table structure

Explanation: TRUNCATE TABLE removes all records from a table but keeps the table and its columns available for future use.

17. Which statement is used to add new records to a table?

- A. ADD
- B. INSERT INTO
- C. UPDATE
- D. CREATE

Answer: B. INSERT INTO

Explanation: INSERT INTO adds one or more new records to an existing table.

18. Which query correctly inserts a student into the following table?

Students

student_id | name | age

- A. INSERT Students VALUES (1, 'Rahul', 20);
- B. INSERT INTO Students VALUES (1, 'Rahul', 20);
- C. ADD INTO Students VALUES (1, 'Rahul', 20);
- D. CREATE INTO Students VALUES (1, 'Rahul', 20);

Answer: B. INSERT INTO Students VALUES (1, 'Rahul', 20);

Explanation: INSERT INTO is the correct SQL syntax for adding records to an existing table.

19. Which SQL command is used to modify existing records?

- A. MODIFY
- B. CHANGE
- C. UPDATE
- D. ALTER

Answer: C. UPDATE

Explanation: UPDATE changes existing data stored in one or more rows.

20. What is the main danger of running the following query?

```
UPDATE Employees  
SET salary = 50000;
```

- A. It deletes all employees
- B. It updates every employee's salary
- C. It creates a new employee
- D. It produces a syntax error

Answer: B. It updates every employee's salary

Explanation: Because there is no WHERE clause, the UPDATE statement applies to every row in the table. A WHERE condition should be used when only specific records need to be changed.

21. Which command is used to remove selected records from a table?

- A. DROP
- B. DELETE
- C. REMOVE
- D. TRUNCATE

Answer: B. DELETE

Explanation: DELETE removes records from a table and can use a WHERE clause to target specific rows.

22. What happens when the following query is executed?

```
DELETE FROM Employees  
WHERE department = 'HR';
```

- A. The entire Employees table is deleted
- B. Only HR employees are deleted
- C. The HR column is deleted
- D. All employees are deleted

Answer: B. Only HR employees are deleted

Explanation: The WHERE clause restricts the DELETE operation to rows whose department is HR.

23. Which statement retrieves data from a table?

- A. GET
- B. SELECT
- C. FETCH TABLE
- D. READ

Answer: B. SELECT

Explanation: SELECT is the primary SQL statement used to retrieve data from one or more tables.

24. What does the following query return?

```
SELECT *  
FROM Employees;
```

- A. Only the first employee
- B. Only the employee names
- C. All columns and all rows from Employees
- D. The structure of Employees only

Answer: C. All columns and all rows from Employees

Explanation: The * symbol represents all columns, while the absence of a WHERE clause means all rows are returned.

25. Which query retrieves only the 'name' and 'salary' columns from an Employees table?

- A. SELECT name, salary FROM Employees;
- B. SELECT Employees FROM name, salary;
- C. GET name, salary FROM Employees;
- D. SELECT * name, salary FROM Employees;

Answer: A. SELECT name, salary FROM Employees;

Explanation: Specific columns can be listed after SELECT, allowing the query to return only the required information instead of every column.

26. Which clause is used to filter records based on a condition?

- A. GROUP BY
- B. WHERE
- C. ORDER BY
- D. HAVING

Answer: B. WHERE

Explanation: The WHERE clause filters individual rows according to a specified condition.

27. What will the following query return?

```
SELECT *  
FROM Students  
WHERE marks > 75;
```

- A. Students with marks less than 75
- B. Students with marks equal to 75 only
- C. Students with marks greater than 75
- D. All students

Answer: C. Students with marks greater than 75

Explanation: The > operator selects only records where the value of marks is greater than 75.

28. Which operator checks whether two values are equal in SQL?

- A. ==
- B. =
- C. ===
- D. !=

Answer: B. =

Explanation: SQL uses a single equals sign (=) for comparison. == is commonly associated with programming languages such as C++ and JavaScript.

29. Which condition selects employees whose salary is less than or equal to 40,000?

- A. salary < 40000
- B. salary <= 40000
- C. salary =< 40000
- D. salary <> 40000

Answer: B. salary <= 40000

Explanation: The <= operator means "less than or equal to," so both 40,000 and values below it are selected.

30. Which operator is used to specify that a value must satisfy both conditions?

- A. OR
- B. NOT
- C. AND
- D. IN

Answer: C. AND

Explanation: AND requires all specified conditions to be true for a record to be selected.

31. What does the following query return?

```
SELECT *  
FROM Employees  
WHERE department = 'IT'  
AND salary > 50000;
```

- A. All IT employees
- B. All employees earning more than 50,000
- C. IT employees earning more than 50,000
- D. IT employees earning exactly 50,000

Answer: C. IT employees earning more than 50,000

Explanation: Both conditions must be true because they are connected using AND.

32. Which operator is used when at least one of multiple conditions should be true?

- A. AND
- B. OR
- C. NOT
- D. BETWEEN

Answer: B. OR

Explanation: OR returns a record when at least one of its conditions evaluates to true.

33. What does the 'NOT' operator do?

- A. Adds two conditions
- B. Sorts records
- C. Reverses a condition
- D. Groups records

Answer: C. Reverses a condition

Explanation: NOT negates a condition. For example, WHERE NOT department = 'HR' selects employees who are not in HR.

34. Which query correctly retrieves customers from either Mumbai or Pune?

- A. SELECT * FROM Customers WHERE city = 'Mumbai' AND city = 'Pune';
- B. SELECT * FROM Customers WHERE city = 'Mumbai' OR city = 'Pune';
- C. SELECT * FROM Customers WHERE city = 'Mumbai' NOT city = 'Pune';
- D. SELECT * FROM Customers WHERE city IN Mumbai, Pune;

Answer: B. SELECT * FROM Customers WHERE city = 'Mumbai' OR city = 'Pune';

Explanation: OR allows either condition to be true. The same requirement can also be written more compactly using IN.

35. Which clause is used to sort query results?

- A. SORT BY
- B. ORDER BY
- C. ARRANGE BY
- D. GROUP BY

Answer: B. ORDER BY

Explanation: ORDER BY sorts the result set according to one or more columns.

36. What is the default sorting direction of 'ORDER BY'?

- A. DESC
- B. RANDOM
- C. ASC
- D. NONE

Answer: C. ASC

Explanation: If neither ASC nor DESC is specified, SQL generally sorts values in ascending order.

37. Which query displays employees from highest salary to lowest salary?

- A. SELECT * FROM Employees ORDER BY salary ASC;
- B. SELECT * FROM Employees SORT salary DESC;
- C. SELECT * FROM Employees ORDER BY salary DESC;
- D. SELECT * FROM Employees GROUP BY salary DESC;

Answer: C. SELECT * FROM Employees ORDER BY salary DESC;

Explanation: DESC sorts values in descending order, placing the highest salaries first.

38. Which query sorts employees first by department and then by salary from highest to lowest within each department?

- A. ORDER BY department, salary DESC
- B. ORDER BY salary, department DESC
- C. GROUP BY department, salary DESC
- D. ORDER BY department DESC salary

Answer: A. ORDER BY department, salary DESC

Explanation: SQL can sort by multiple columns. The first column establishes the primary ordering, while the second controls ordering within matching values.

39. Which clause is used to restrict the number of rows returned by a MySQL query?

- A. TOP
- B. LIMIT
- C. RESTRICT
- D. FIRST

Answer: B. LIMIT

Explanation: MySQL uses LIMIT to restrict the number of rows returned by a query.

40. What will this query return?

```
SELECT *  
FROM Products  
LIMIT 10;
```

- A. Exactly 10 columns
- B. At most 10 rows
- C. Exactly 10 databases
- D. The last 10 rows

Answer: B. At most 10 rows

Explanation: LIMIT 10 restricts the result to a maximum of 10 rows. If fewer than 10 records exist, fewer rows will be returned.

41. Which keyword is used to give a temporary name to a column or table?

- A. NAME
- B. AS
- C. RENAME
- D. LABEL

Answer: B. AS

Explanation: The AS keyword creates an alias for a column or table in the query result.

42. What will the following query do?

```
SELECT  
salary AS Monthly_Salary  
FROM Employees;
```

- A. Rename the actual database column permanently
- B. Create a new salary column
- C. Display salary using the alias Monthly_Salary
- D. Delete the salary column

Answer: C. Display salary using the alias Monthly_Salary

Explanation: A column alias changes how the column is displayed in the query result. It does not permanently rename the database column.

43. Which keyword is used to remove duplicate values from a result?

- A. UNIQUE
- B. DISTINCT
- C. REMOVE
- D. DIFFERENT

Answer: B. DISTINCT

Explanation: DISTINCT returns only unique combinations of the selected columns.

44. What does the following query return?

```
SELECT DISTINCT city  
FROM Customers;
```

- A. Every customer record
- B. Only unique cities
- C. Only duplicate cities
- D. The number of cities

Answer: B. Only unique cities

Explanation: DISTINCT eliminates duplicate city values from the result.

45. Which query correctly displays the 5 highest-paid employees?

- A. SELECT * FROM Employees LIMIT 5 ORDER BY salary DESC;
- B. SELECT * FROM Employees ORDER BY salary DESC LIMIT 5;
- C. SELECT * FROM Employees ORDER salary LIMIT 5;
- D. SELECT * FROM Employees GROUP BY salary LIMIT 5;

Answer: B. SELECT * FROM Employees ORDER BY salary DESC LIMIT 5;

Explanation: The results must first be sorted in descending order and then limited to five rows.

46. Which wildcard represents any number of characters with 'LIKE'?

- A. _
- B. %
- C. *
- D. #

Answer: B. %

Explanation: In SQL pattern matching, % represents zero or more characters.

47. Which wildcard represents exactly one character with 'LIKE'?

- A. %
- B. _
- C. *
- D. ?

Answer: B. _

Explanation: The underscore (_) wildcard represents exactly one character in a LIKE pattern.

48. What does the following query find?

```
SELECT *  
FROM Customers  
WHERE name LIKE 'A%';
```

- A. Names ending with A
- B. Names containing exactly one A
- C. Names beginning with A
- D. Names containing no A

Answer: C. Names beginning with A

Explanation: A% means the value must begin with A, followed by zero or more characters.

49. Which query finds cities containing the word 'park' anywhere in the city name?

- A. WHERE city LIKE 'park%'
- B. WHERE city LIKE '%park'
- C. WHERE city LIKE '%park%'
- D. WHERE city = 'park'

Answer: C. WHERE city LIKE '%park%'

Explanation: % before and after park allows any number of characters to appear on either side of the word.

50. What does the following query find?

```
SELECT *  
FROM Products  
WHERE product_name LIKE 'C__';
```

- A. Products beginning with C and having exactly three characters
- B. Products containing C anywhere
- C. Products ending with C
- D. Products beginning with C and having at least three characters

Answer: A. Products beginning with C and having exactly three characters

Explanation: C__ consists of C followed by exactly two characters. Therefore, the complete value must contain exactly three characters and start with C.

51. Which SQL operator is used to check whether a value falls within a specified range?

- A. IN
- B. BETWEEN
- C. RANGE
- D. WITHIN

Answer: B. BETWEEN

Explanation: BETWEEN checks whether a value lies within an inclusive range. For example, WHERE salary BETWEEN 30000 AND 50000 includes both 30,000 and 50,000.

52. What will the following query return?

```
SELECT *  
FROM Employees  
WHERE salary BETWEEN 30000 AND 50000;
```

- A. Employees earning less than 30,000
- B. Employees earning more than 50,000
- C. Employees earning from 30,000 through 50,000, including both limits
- D. Employees earning exactly 30,000 or 50,000 only

Answer: C. Employees earning from 30,000 through 50,000, including both limits

Explanation: BETWEEN is inclusive, so values equal to both the lower and upper boundaries are included.

53. Which operator is useful for checking whether a value matches any value from a specified list?

- A. BETWEEN
- B. IN
- C. MATCH
- D. LIST

Answer: B. IN

Explanation: IN allows a column to be compared against multiple specified values without writing separate OR conditions.

54. Which query correctly retrieves employees from the IT, HR, or Sales departments?

- A. WHERE department IN ('IT', 'HR', 'Sales')
- B. WHERE department BETWEEN ('IT', 'HR', 'Sales')
- C. WHERE department LIKE ('IT', 'HR', 'Sales')
- D. WHERE department = IN ('IT', 'HR', 'Sales')

Answer: A. WHERE department IN ('IT', 'HR', 'Sales')

Explanation: IN checks whether a value matches any value in the supplied list.

55. Which condition checks whether a column contains a NULL value?

- A. column = NULL
- B. column IS NULL
- C. column == NULL
- D. column NULL

Answer: B. column IS NULL

Explanation: SQL uses IS NULL to test for missing or NULL values. Using = NULL does not correctly test for NULL.

56. Which condition finds records where an email address has been provided?

- A. email != NULL
- B. email IS NOT NULL
- C. email <> NULL
- D. email EXISTS

Answer: B. email IS NOT NULL

Explanation: IS NOT NULL returns rows where the specified column contains a value rather than NULL.

57. Why should '= NULL' generally not be used to check for missing values?

- A. NULL is always equal to zero
- B. NULL represents an unknown or missing value and requires special comparison operators
- C. NULL can only be used with numbers
- D. SQL automatically converts NULL to an empty string

Answer: B. NULL represents an unknown or missing value and requires special comparison operators

Explanation: NULL represents the absence or unknown state of a value. SQL therefore uses IS NULL and IS NOT NULL for NULL comparisons.

58. Which query finds customers whose city starts with "M" and whose state is "MH"?

- A. WHERE city LIKE 'M%' AND state = 'MH'
- B. WHERE city LIKE '%M' OR state = 'MH'
- C. WHERE city = 'M%' AND state LIKE 'MH'
- D. WHERE city LIKE 'M_' OR state = 'MH'

Answer: A. WHERE city LIKE 'M%' AND state = 'MH'

Explanation: M% matches cities beginning with M, while AND requires state condition to also be true.

59. Which clause should be used to filter records before grouping?

- A. HAVING
- B. WHERE
- C. GROUP BY
- D. ORDER BY

Answer: B. WHERE

Explanation: WHERE filters individual rows before grouping and aggregation take place.

60. Which clause is specifically designed to filter grouped or aggregated results?

- A. WHERE
- B. ORDER BY
- C. HAVING
- D. DISTINCT

Answer: C. HAVING

Explanation: HAVING filters groups after GROUP BY has been applied and is commonly used with aggregate functions.

61. Which function returns the number of rows in a table?

- A. SUM()
- B. COUNT()
- C. TOTAL()
- D. NUMBER()

Answer: B. COUNT()

Explanation: COUNT() counts rows or non-NULL values depending on how it is used. COUNT(*) counts all rows.

62. What does the following query return?

```
SELECT COUNT(*)  
FROM Employees;
```

- A. The total number of columns
- B. The total number of employees/rows
- C. The sum of employee salaries
- D. The number of unique employees only

Answer: B. The total number of employees/rows

Explanation: COUNT(*) counts every row in the table, regardless of whether individual columns contain NULL values.

63. Which aggregate function calculates the total of numeric values?

- A. COUNT()
- B. AVG()
- C. SUM()
- D. TOTAL()

Answer: C. SUM()

Explanation: SUM() adds together the values of a numeric column.

64. Which aggregate function calculates the average value?

- A. MEAN()
- B. AVG()
- C. AVERAGE()
- D. MID()

Answer: B. AVG()

Explanation: AVG() calculates the arithmetic mean of the numeric values in a column.

65. Which function returns the smallest value in a column?

- A. LOW()
- B. MIN()
- C. SMALL()
- D. LEASTVALUE()

Answer: B. MIN()

Explanation: MIN() returns the smallest value from the selected column or expression.

66. Which function returns the largest value in a column?

- A. MAX()
- B. HIGH()
- C. LARGEST()
- D. TOP()

Answer: A. MAX()

Explanation: MAX() returns the largest value found in the selected column.

67. What will this query calculate?

```
SELECT AVG(salary)
FROM Employees;
```

- A. Highest salary
- B. Lowest salary
- C. Average salary
- D. Total salary

Answer: C. Average salary

Explanation: AVG() calculates the arithmetic average of the salary values.

68. Which clause is used to create groups of rows having the same value?

- A. GROUP BY
- B. ORDER BY
- C. PARTITION BY
- D. ARRANGE BY

Answer: A. GROUP BY

Explanation: GROUP BY combines rows with the same values in specified columns so aggregate functions can be applied to each group.

69. What will the following query produce?

```
SELECT department, COUNT(*)  
FROM Employees  
GROUP BY department;
```

- A. One total count for the entire table
- B. Number of employees in each department
- C. Only departments with one employee
- D. Employee names sorted by department

Answer: B. Number of employees in each department

Explanation: GROUP BY department creates one group for each department, and COUNT(*) counts the employees within each group.

70. Which query correctly finds departments having more than 10 employees?

- A. SELECT department, COUNT(*) FROM Employees WHERE COUNT(*) > 10 GROUP BY department;
- B. SELECT department, COUNT(*) FROM Employees GROUP BY department HAVING COUNT(*) > 10;
- C. SELECT department, COUNT(*) FROM Employees GROUP BY department WHERE COUNT(*) > 10;
- D. SELECT department FROM Employees HAVING COUNT(*) > 10;

Answer: B. SELECT department, COUNT(*) FROM Employees GROUP BY department HAVING COUNT(*) > 10;

Explanation: Aggregate results cannot normally be filtered using WHERE. HAVING is used to filter groups based on aggregate calculations.

71. What is the main difference between WHERE and HAVING?

- A. WHERE sorts data, while HAVING groups data
- B. WHERE filters rows, while HAVING filters groups
- C. WHERE creates tables, while HAVING deletes tables
- D. There is no difference

Answer: B. WHERE filters rows, while HAVING filters groups

Explanation: WHERE operates on individual records before grouping, while HAVING operates on grouped results after aggregation.

72. Which query calculates total payment amount for each payment type?

- A. SELECT payment_type, SUM(payment_value) FROM Payments GROUP BY payment_type;
- B. SELECT payment_type, TOTAL(payment_value) FROM Payments;
- C. SELECT payment_type, SUM(payment_value) FROM Payments ORDER BY payment_type;
- D. SELECT SUM(payment_type), payment_value FROM Payments GROUP BY payment_value;

Answer: A. SELECT payment_type, SUM(payment_value) FROM Payments GROUP BY payment_type;

Explanation: GROUP BY payment_type creates a group for each payment method, and SUM() calculates the total payment value for each group.

73. What does the following query return?

```
SELECT department, MAX(salary)
FROM Employees
GROUP BY department;
```

- A. The highest salary across the entire company
- B. The lowest salary in each department
- C. The highest salary in each department
- D. Every employee's salary

Answer: C. The highest salary in each department

Explanation: Each department becomes a separate group, and MAX(salary) returns the highest salary within each group.

74. Which statement about aggregate functions is correct?

- A. They can only be used with text columns
- B. They summarize multiple values into a result
- C. They always return every row
- D. They can only be used with WHERE

Answer: B. They summarize multiple values into a result

Explanation: Aggregate functions such as COUNT(), SUM(), AVG(), MIN(), and MAX() perform calculations across multiple rows.

75. Which query correctly finds payment types where the average payment is greater than 500?

- A. SELECT payment_type, AVG(payment_value) FROM Payments WHERE AVG(payment_value) > 500 GROUP BY payment_type;
- B. SELECT payment_type, AVG(payment_value) FROM Payments GROUP BY payment_type HAVING AVG(payment_value) > 500;
- C. SELECT payment_type, AVG(payment_value) FROM Payments HAVING payment_value > 500;
- D. SELECT payment_type FROM Payments WHERE payment_value > AVG(500);

Answer: B. SELECT payment_type, AVG(payment_value) FROM Payments GROUP BY payment_type HAVING AVG(payment_value) > 500;

Explanation: The average must first be calculated for each payment type using GROUP BY. HAVING then filters those groups based on the calculated average.